



AWS AppSync

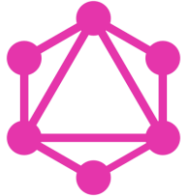
Lecture Slides

These lecture slides are provided for personal and non-commercial use only

Please do not redistribute or upload these lecture slides elsewhere.

Good luck on your exam!

GraphQL



GraphQL is an **open-source agnostic query adaptor** that allows you to query data from many different data sources.

GraphQL is used to build APIs where clients will send a query for nested data.

GraphQL mitigates the issue of versioned or rapidly changing APIs compared to REST API because you can request the data you want.

GraphQL schemas are written in the GraphQL SDL (Schema Definition Language) composed of:

- Types
- Fields
- Queries
- Mutations
- Subscriptions

Type

Represent objects and their fields

```
type Query {  
  me: User  
}  
  
type User {  
  id: ID  
  name: String  
}
```

Queries

defines the exact shape of the data needed by the client.

```
query CurrentUser {  
  currentUser {  
    name  
    age  
  }  
}
```

Mutations

allows for data to be created, updated, or deleted.

```
mutation CreateUser($name: String!, $age: Int!) {  
  createUser(userName: $name, age: $age) {  
    name  
    age  
  }  
}
```

Subscriptions

supports live updates sent from the server to client

```
subscription {  
  newPerson {  
    name  
    age  
  }  
}
```

AWS AppSync



AppSync is a **fully managed GraphQL service**.

Amplify SDK



AppSync

Data Model

```
{
  "data": {
    // resolves to...
    "project": {
      "id": 1,
      "name": "My project"
    },
    // resolves to...
    "tasks": [{
      "id": 1,
      "name": "My todo"
    }]
  }
}
```

Resolvers

attached to specific fields within your types in your schema.
used to implement the state-changing operations for your query, mutation, and subscription field operations.

Resolver (Project)



Resolver (Task)



Data Sources

Where data is being pulled from

AWS AppSync can support real-time APIs via GraphQL subscriptions feature

AWS AppSync

API Types:

- **GraphQL APIs** — single API from multiple data sources
- **Merged APIs** — a collection of Graph APIs that act as one API
 - Useful if you have multiple teams that manage their own API but you want to treat all the APIs a single API

Data Sources:

- DynamoDB Table
- Amazon OpenSearch
- AWS Lambda Function
- HTTP Endpoint
- Amazon EventBridge
- Relational Database (RDS HTTP Endpoint)

Caching options:

- **None** — no caching, resolvers will always fetch from data sources
- **Full request caching** — Cache all requests
- **Per-resolver caching** — specific operation or field defined in a resolver will return responses from the cache

Caching required you to provision an on-demand instance. Its not serverless.

Authorization Types:

- API Key
- AWS IAM
- Amazon Cognito User Pools

- AppSync supports custom domains
- AppSync has a query editor built into the UI

Resolver Runtimes

- **APPSYNC_JS** — Javascript
- **VTL** — Velocity Template Language